

Experiment- 7

Concave lens

AIM:

To find the focal length of a concave lens using a convex lens.

APPARATUS:

An optical bench, a concave lens, a convex lens, two lens holders, two optical needles, a knitting needle, and a half-meter scale.

THEORY:

The focal length F of the combination (convex lens and concave lens) is given by:

$$\frac{1}{F} = \frac{1}{f_1} - \frac{1}{f_2}$$

Where f_1 is the focal length of the convex lens and f_2 is the focal length of the concave lens

Using the lens equation,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Where u is the object distance, and v is the image distance.

PROCEDURE:

1. Take the rough focal length of the convex lens f_1 by the distant object method. Twice the value of the focal length gives $2f$ of the given lens.
2. Obtain the rough focal length of the combination (convex lens and concave lens) by focusing the image of a distant object on the wall. Twice the value of the focal length gives $2f$ of the given lens.
3. Mount the given lens (combination) on the lens holder and place it upright in the appropriate position of an optical bench, which is being leveled with the help of leveling screws and a spirit level.
4. Place the mounted pin (object needle P_1) from the combination more than twice its focal length.
5. Obtain its real and inverted image I on the other side of the lens by removing the parallax between the image and the second needle P_2 . Note the position of P_2 on the optical bench.
6. Let the distance of the object needle from the combination be u , and the distance of the image needle from the combination be v .
7. Now move the object needle 1 or 2 cm in front of the combination.

8. Repeat the above steps. Note the values of u and v .

RESULT:

The focal length of the given concave lens =

Precautions

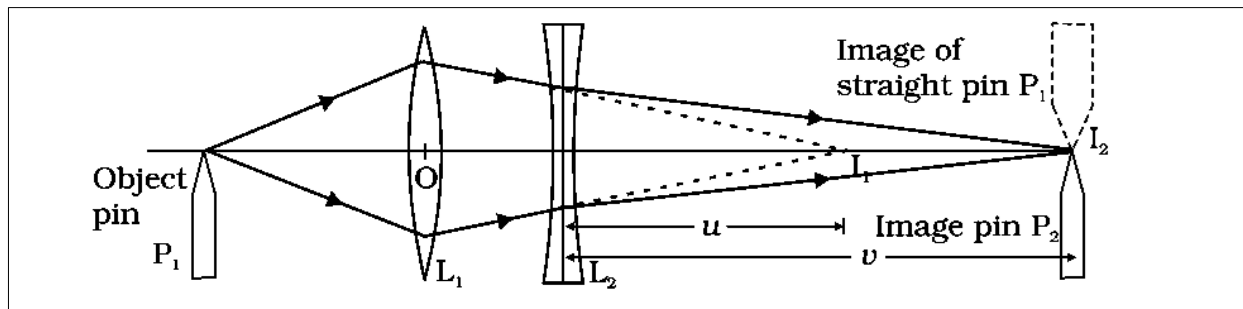
- The focal length of the convex lens should be less than the focal length of the concave lens so that the combination is convex.
- The lenses must be clean.

Sources of Error

- The uprights may not be vertical.
- Parallax removal may not be perfect.

Diagram and observation table on the left side of the page(diagram on one page and observation tables on the other page)

Ray Diagram:



Observation table

Rough focal length of the convex lens, $f_1 =$

Rough focal length of the combination =

S.No	$u(\text{cm})$	$v(\text{cm})$	$F = \frac{uv}{u+v} (\text{cm})$
1			
2			
3			
4			

Calculations:

Mean $F =$

$$f_2=\frac{f_{1F}}{f_1-F}=$$